

E-03: Bounded enstrophy implies continuation - PROVEN

NS-BLOWUP-NECESSARY

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Estado

$$\boxed{E-03 = \text{PROVEN}}$$

Marco

Leray-Hopf en $\mathbb{R}^3$, $u_0 \in H^1$, $\text{div } u_0 = 0$, rama fuerte H^1 en $[0, T_*)$, T_* maximal.

$$\|\nabla u(t)\|_2 = \|\omega(t)\|_2.$$

Cadena

$$\sup_{t < T_*} \|\omega\|_2 < \infty \Downarrow \sup_{t < T_*} \|\nabla u\|_2 < \infty \Downarrow u \in L^\infty(0, T_*; L^6)$$

$$\|u\|_{L^6} \leq C_S \|\nabla u\|_2, \quad p = \infty, q = 6, \quad \frac{2}{p} + \frac{3}{q} = \frac{1}{2} < 1.$$

$$\boxed{u \in L^\infty(0, T_*; L^6) \implies \text{Serrin subcritico } (\infty, 6)}$$

$$\boxed{\text{Serrin} \implies u \text{ continuable mas alla de } T_*}.$$

Si $T_* < \infty$ maximal, esto es $\perp$.

$$\boxed{\sup_{t < T_*} \|\omega(t)\|_2 < \infty \implies T_* = \infty}.$$

$$\boxed{T_* < \infty \implies \sup_{t < T_*} \|\omega\|_2 = \infty}.$$